MTH632 Quiz Solved

① $f(z) = x + ay + i(bx + cy)$

$u(x,y) = x + ay$, $v(x,y) = bx + cy$

$u_x = 1$, $u_y = a$, $v_x = b$, $v_y = c$

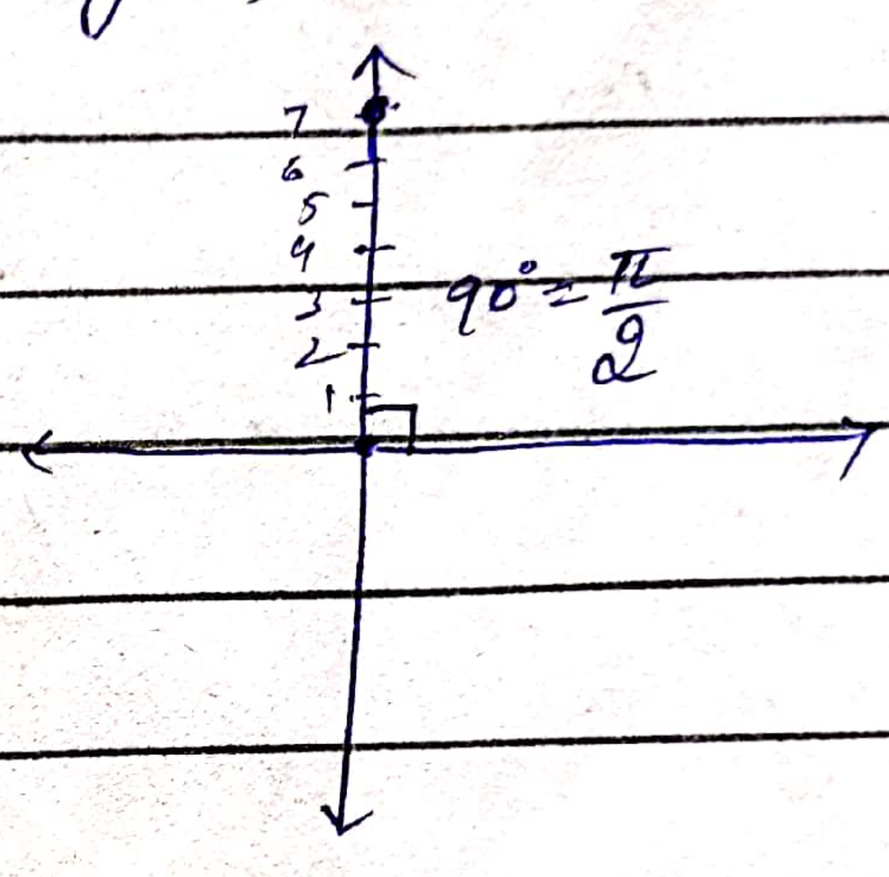
By Reimann equations

$u_x = v_y$	$u_y = -v_x$
$1 = c$	$a = -b$
$\boxed{c = 1}$	$\boxed{a = -b}$

② let $z = 7i$, the $\arg(z) =$ ——

$z = 0 + 7i$

$x = 0$, $y = 7$



$\arg(z) = 90° = \pi/2$

③ If $f(z) = z^3$, $g(z) = z + 2$, then $g(f(4)) = ?$

$f(4) = (4)^3 = 64$

$g(z) = z + 2$

$g(f(4)) = 64 + 2 = 66$

④ $f(x) = x^3$, $g(x) = x + 2$, $g(f(5)) = ?$

$f(5) = 5^3 = 125$

$g(f(5)) = 125 + 2 = 127$

Scanned with CamScanner

(5)

$z = x - iy$

$f(z) = \bar{z} = \overline{x - iy} = x + iy$

$f(z) = x + iy$

$u(x,y) = x$, $v(x,y) = y$

$\boxed{u_x = 1}$, $\boxed{u_y = 0}$ | $\boxed{v_x = 0}$, $\boxed{v_y = 1}$

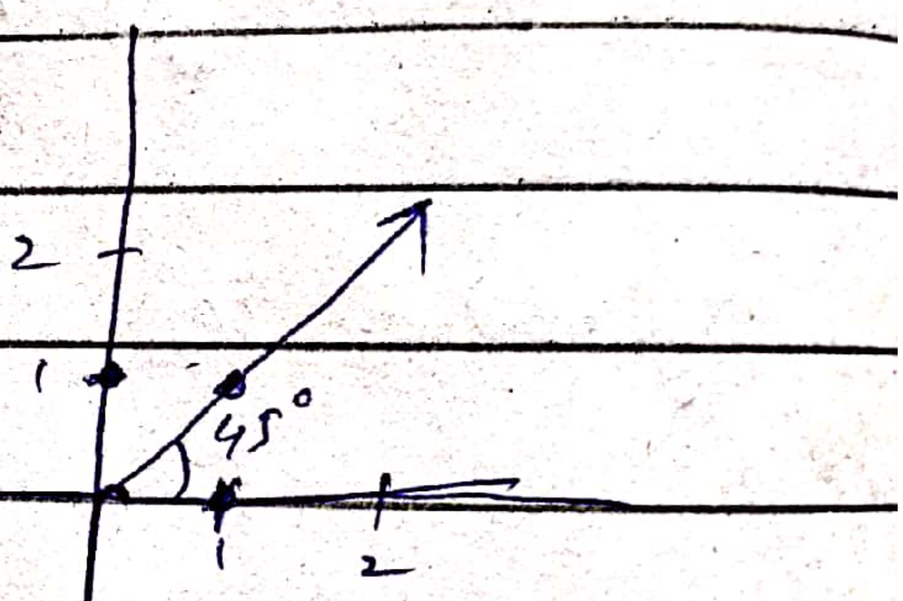
$u_x = v_y$ | $u_y = -v_x$

Hence cauchy Reimann eq are satisfied

(6) let $z = 1 + i$, the $\arg(z) =$

$x = 1 , y = 1$

$\arg(z) = 45° = \frac{\pi}{4}$



(7) Harmonic conjugate of $u(x,y) = e^y \cos x$ is =

$u(x,y) = e^y \cos x$

$u_x = -e^y \sin x$, $u_y = e^y \cos x$

By Reimann eq: $u_x = v_y \rightarrow u_y = -v_x$

$v_y = -e^y \sin x$, $-v_x = e^y \cos x$

$\Rightarrow v_x = -e^y \cos x$

$\int v_y \, dy = -\int e^y \sin x \, dy =$

$= -e^y \sin x + c$

$\int v_x \, dx = -\int e^y \cos x \, dx$

$= -e^y \sin x + c$

So $\boxed{v(x,y) = -e^y \sin x + c}$

Scanned with CamScanner

(8) A function V is said to be harmonic if and only if $\boxed{V_{xx} + V_{yy} = 0}$

(9) If $f(z) = \dfrac{z^4+1}{(z^2+1)(z-3)}$, then $f(z)$ is not analytic at $z =$

put $(z^2+1)(z-3) = 0$

$z^2+1 = 0$, $z-3 = 0$

$z^2 = -1$, $\boxed{z = 3}$

$z = \pm\sqrt{-1}$,

$\boxed{z = \pm i}$

$f(z)$ is not analytic at $z = \pm i, 3$

(10) Sum of complex numbers $(3+2i)$ and $(1+7i)$ is

Sum $= (3+2i)+(1+7i) = 3+2i+1+7i$

$= 4+9i$

(11) Conjugate of complex number $8+3i$ is $\boxed{8-3i}$

(12) $f(z) = z^2$, $g(z) = \dfrac{1}{z}$, then $g(f(4)) = ?$

$f(4) = 4^2 = 16$

$g(f(4)) = \dfrac{1}{f(4)} = \dfrac{1}{16}$

Scanned with CamScanner

(13) Sum of complex numbers $(3+5i)$ and $(4-3i)$ is

$$\text{Sum} = 3+5i+4-3i$$

$$\text{Sum} = 7+2i$$

(14) let $z = 1+i$, the $r = ?$

$$x = 1, \quad y = 1$$

$$r = \sqrt{x^2+y^2} = \sqrt{1^2+1^2} = \sqrt{1+1} = \sqrt{2}$$

(15) The cauchy - Reimann equations.

$$\boxed{U_x = V_y} \quad \boxed{U_y = -V_x}$$

(16) If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(i)) =$

$$f(i) = i^3 = i^2 \cdot i = -i$$

$$g(z) = \frac{1}{z}$$

$$g(f(i)) = \frac{1}{f(i)} = \frac{1}{-i} \times \frac{i}{i} = \frac{i}{-i^2}$$

$$= \frac{i}{-(-1)} = \frac{i}{1} = i$$

(17) let $z = 7i$, then $r = ?$

$$x = 0, \quad y = 7$$

$$r = \sqrt{x^2+y^2} = \sqrt{0^2+7^2} = \sqrt{49} = 7$$

(18) A curve is called [simple curve] if it does not cross itself.

Scanned with CamScanner

(19) $(1+i)^2 = ?$

$(1+i)^2 = (1)^2 + (i)^2 + 2(1)(i)$

$= 1 + i^2 + 2i$

$= 1 - 1 + 2i$

$= 2i$

$\boxed{(1+i)^2 = 0 + 2i}$

(20) If $e^{ax}\cos y$ is harmonic then "a" is —

let $f(x,y) = e^{ax}\cos y$

$\frac{\partial f}{\partial x} = ae^{ax}\cos y$

$\frac{\partial^2 f}{\partial x^2} = a^2 e^{ax}\cos y$

$\frac{\partial f}{\partial y} = -e^{ax}\sin y$

$\frac{\partial^2 f}{\partial y^2} = -e^{ax}\cos y$

Since $f(x,y)$ is harmonic, we have

$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$

$\Rightarrow a^2 e^{ax}\cos y - e^{ax}\cos y = 0$

$\Rightarrow e^{ax}\cos y (a^2 - 1) = 0$

$e^{ax}\cos y = 0$ and $a^2 - 1 = 0$

The value of a is ± 1 $\quad \boxed{a = \pm 1}$

Scanned with CamScanner

(21) If $f(z) = z^2$, $g(z) = \frac{1}{z}$, the $g(f(3)) =$

$f(3) = (3)^2 = 9$

$g(z) = \frac{1}{z}$

$g(f(3)) = \frac{1}{f(3)} = \frac{1}{9}$

(22) If $f(z) = \frac{z^4+1}{(z^2+1)(z-5)}$, then $f(z)$ is not analytic at $z =$

put $(z^2+1)(z-5) = 0$

$z^2+1 = 0$, $z-5 = 0$

$z = \pm i$, $z = 5$

(23) If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(-1)) =$

$f(-1) = (-1)^3 = -1$

$g(f(-1)) = \frac{1}{f(-1)} = \frac{1}{-1} = -1$

(24) A function $f(z)$ is analytic if real and imaginary part $f(z)$ is analytic.

(25) A point at which a function ceases to be analytic is called a singular point.

Scanned with CamScanner

(26) $f(z) = z^2$, then $f(x+iy) = ?$

$$f(z) = z^2$$

$$f(x+iy) = (x+iy)^2$$

$$= (x)^2 + (iy)^2 + 2(x)(iy)$$

$$= x^2 + i^2y^2 + 2xyi$$

$$f(x+iy) = x^2 - y^2 + 2xyi$$

(27) $$\lim_{(x,y)\to(1,2)} (3xy^2 - y) =$$

$$\lim_{(x,y)\to(1,2)} (3xy^2 - y) = 3(1)(2)^2 - (2)$$

$$= 12 - 2$$

$$= 10$$

(28) product of complex numbers $(3+2i)$ and $(1+7i)$ is

$$\text{product} = (3+2i)(1+7i)$$

$$= (3)(1) + (3)(7i) + (2i)(1) + (2i)(7i)$$

$$= 3 + 21i + 2i + 14i^2$$

$$= 3 + 23i - 14$$

$$= -11 + 23i$$

(29) The function $u(x,y) = \frac{xy}{x^2+y^2}$ does not have a limit $\boxed{(x,y) \to (0,0)}$

Because function $u(x,y)$ undefined on

(30)

If $-8i = 8e^{i\left(\frac{-\pi}{2}+2k\pi\right)}$ then the root of $(-8i)^{1/3}$

$$\Rightarrow (-8i)^{1/3} = \left[8e^{i\left(\frac{-\pi}{2}+2k\pi\right)}\right]^{1/3}$$

$$= (2^3)^{1/3} e^{i\left(\frac{-\pi}{2}\times\frac{1}{3} + 2k\pi\times\frac{1}{3}\right)}$$

$$= 2e^{i\left(\frac{-\pi}{6} + \frac{2k\pi}{3}\right)}$$

$$(-8i)^{1/3} = 2e^{i\left(\frac{-\pi+6k\pi}{6}\right)}$$

(1) For $K = 0$

$$(-8i)^{1/3} = 2e^{i\left(\frac{-\pi+0}{6}\right)}$$

$$= 2e^{i\left(\frac{-\pi}{6}\right)}$$

$$= 2\left(\cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right)\right)$$

$$= 2\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$$

$$= 2\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right)$$

$$= 2\left(\frac{\sqrt{3}-i}{2}\right)$$

For $K=0$ $\boxed{(-8i)^{1/3} = \sqrt{3} - i}$

Scanned with CamScanner

(31) (2) For $k=1$

$$(-8i)^{1/3} = 2e^{i\left(\frac{-\pi+4\pi}{6}\right)}$$

$$= 2e^{i\left(\frac{3\pi}{6}\right)}$$

$$= 2e^{i\left(\frac{\pi}{2}\right)}$$

$$= 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

$$= 2(0 + i(1))$$

For $k=1$ $\boxed{(-8i)^{1/3} = 2i}$

(32) (3) For $k=2$

$$(-8i)^{1/3} = 2e^{i\left(\frac{-\pi+8\pi}{6}\right)}$$

$$= 2e^{i\left(\frac{7\pi}{6}\right)}$$

$$= 2\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$$

$$= 2\left(\frac{-\sqrt{3}}{2} + i\left(\frac{1}{2}\right)\right)$$

$$= 2\left(\frac{-\sqrt{3}}{2} - \frac{1}{2}i\right)$$

$$= 2\left(\frac{-\sqrt{3}-i}{2}\right)$$

For $k=2$ $\boxed{(-8i)^{1/3} = -\sqrt{3} - i}$

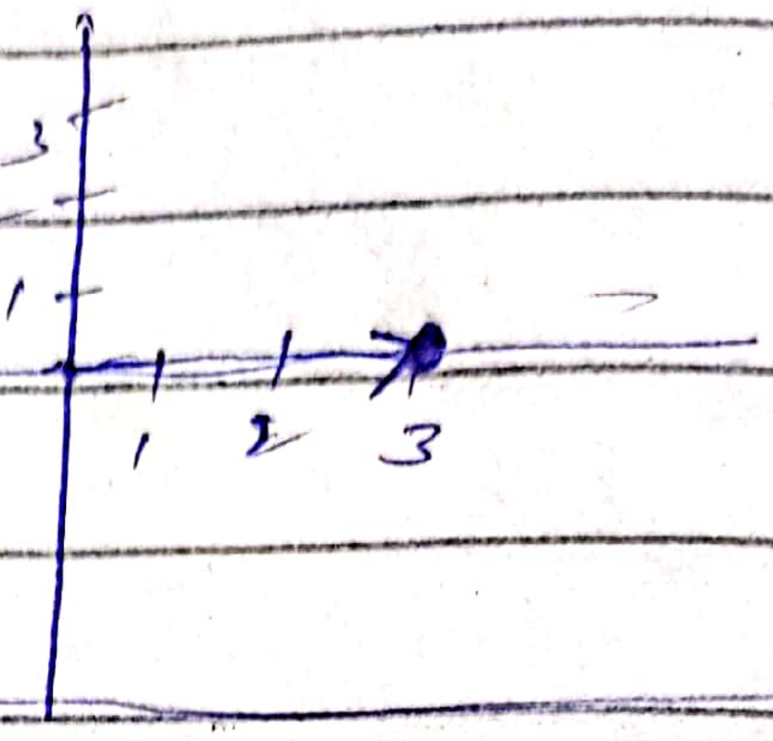
Scanned with CamScanner

(33)

Let $z = 3$, the $\arg(z) = ?$

$x = 3, y = 0$

$\arg(z) = 0^\circ$



OR

$\arg(z) = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(0)$

$= 0^\circ$

(34)

The value $w = \frac{1}{z}$, when $z = 2 - 2i$ is

$$w = \frac{1}{z} = \frac{1}{2-2i} \times \frac{2+2i}{2+2i}$$

$$= \frac{2+2i}{(2)^2 - (2i)^2} = \frac{2+2i}{4-(-1)}$$

$$= \frac{2+2i}{5} = \frac{1}{5}(2+2i)$$

(35)

$w = z^2 - z$ as $w = u(x,y) + iv(x,y)$, then $u(x,y) =$

$w = z^2 - z$

$w = (x+iy)^2 - (x+iy)$

$= x^2 + y^2 + 2xyi - x - iy$

$w = (x^2 - y^2 - x) + i(2xy - y)$

$w = u(x,y) + iv(x,y)$

so $u(x,y) = x^2 - y^2 - x$

and $v(x,y) = 2xy - y$

Scanned with CamScanner

Domain of the Functions

(36) $f(z) = \frac{1}{z^2+1}$, Ans $\boxed{z \neq \pm i}$

(37) $f(z) = \frac{1}{z^2-1}$, Ans $\boxed{z \neq \pm 1}$

(38) $f(z) = \frac{1}{z^2}$, Ans $\boxed{z = 0}$

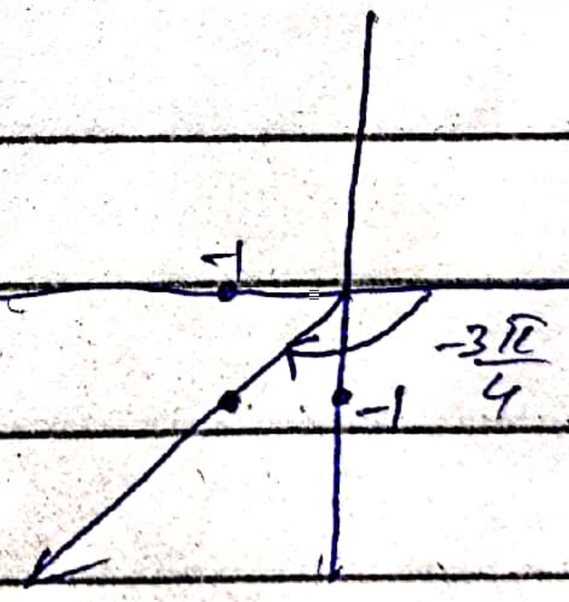
(39) $f(z) = \frac{1}{z+\bar{z}}$, Ans $\boxed{z \neq \pm i}$

(40) The complex number $z = -1-i$ has exponential form

$z = -1-i$

$x = -1, \quad y = -1$

$\arg(z) = \theta = \frac{-3\pi}{4}$



$r = \sqrt{x^2+y^2} = \sqrt{(-1)^2+(-1)^2} = \sqrt{1+1} = \sqrt{2}$

$z = re^{i\theta}$

$-1-i = \sqrt{2}e^{i\left(\frac{-3\pi}{4}\right)}$

is exponential form

Solved By Mahar Azaq Safdar Muhammadi

Scanned with CamScanner